SECTIONC DESCRIPTION/SPECS./WORK STATEMENT DRAFT

C.1 INTRODUCTION

The National Highway Traffic Safety Administration's (NHTSA) mission is to save lives, prevent injuries and reduce traffic-related health care and other economic costs. The agency develops, promotes and implements effective educational, engineering and enforcement programs with the goal of ending vehicle crash tragedies and reducing economic costs associated with vehicle use and highway travel. The Office of Vehicle Safety Research conducts extensive research, development, testing, crash investigation, and data collection and analysis activities to provide the scientific basis needed to support the Agency's motor vehicle and traffic safety goals.

The Office of Vehicle Safety Research (VSR) is the primary organization within NHTSA responsible for conducting research related to the improvement of motor vehicle performance to enhance safety. VSR conducts research and development programs to develop and advance the scientific knowledge to support NHTSA's mission.

C.2 SCOPE OF WORK

As part of this research, VSR develops and exercises crash simulation models to evaluate potential crash conditions and their effects on the safety response of vehicles and human surrogates (dummies) subject to vehicle-to-vehicle, vehicle-to-barrier, and other impacts. Also, as part of this research, VSR develops interior and restraint models of vehicles and performs different types of simulations with different dummies and human models to simulate real-world situations.

C.3 OBJECTIVE

The objective of this contract is to provide research data so that NHTSA can achieve its mission of reducing death and injuries on American roadways. The scope of this contract is to develop and exercise crash simulations of vehicle safety research tests to assist in predict the impact of potential safety implications. These efforts will include full vehicle laser scanning and tear down for the development of new finite element models, material testing to define the behavior of vehicle materials under impact conditions, mesh development, and part integration to develop full vehicle and vehicle component finite element models. Some tasks may involve the use of lumped parameter modeling software such as MADYMO, but majority of NHTSA's anticipated crash simulation efforts will involve non-linear finite element analysis. The resulting models will be exercised and refined to match available crash test data. The resulting LS-Dyna models shall be exercised using existing vehicle models in LS-Dyna simulations. The specific crash simulation efforts will be issued through Task Orders.

C.4 KICK OFF MEETING

Within four (4) weeks of the contract award, the Contractor shall meet with the Contracting Officer's Representative (COR) and interested NHTSA staff via teleconference. The purpose of this meeting is to discuss the contract objectives, the contract administration and the ordering procedures for issuing future Task Orders. The Contractor shall assume no more than 2 hours for this meeting.

C.5 POTENTIAL TASK ORDER REQUIREMENTS

If, during the performance of this Contract, the Government decides to issue Task Orders, they will be based on the potential task orders detailed below. At contract award, however, the Contractor(s) shall not perform work associated with said Tasks below as part of the base requirement above (C.4 Task 1: Kick-Off Meeting). The Contractor(s) will only be required to perform Tasks if the government issues Task Orders in accordance with Section G.6, Task Order Procedures.

C.5.1 The Contractor shall provide the facilities, personnel, expertise, material, and equipment necessary to perform the proposed computer simulations. These simulations should include replication of staged vehicle crash test scenarios and procedures for objective comparison against staged laboratory test results.

C.5.2 The Contractor shall be able to conduct a full vehicle tear down and digitization for the development of crash simulation models suitable for vehicle safety research applications.

C.5.3 The Contractor shall be able to conduct a material component tests and analysis of results suitable for inclusion into crash simulation models. These capabilities should include the characterization of metals, plastics, foams, and composites. Material testing may also include the characterization of welds and other material joining techniques. Some testing may also involve evaluating bulk material characteristics such as force-deflection response for foams and padding materials.

C.5.4 The Contractor shall be able to develop and utilize the simulation models for vehicle interiors and occupant restraint systems. Restraint components such as seats, seat belts, seat belt retractors, air bags, knee bolsters and appropriately positioning dummy models to evaluate the safety performance.

C.5.5 The Contractor shall be able to conduct optimization studies to evaluate the safety effects for a range of vehicle structural designs or occupant restraint characteristics.

C.5.6 The Contractor shall be able to compare simulation results against field reported safety data. NHTSA is seeking support for objective and quantitative comparisons between test and simulation data.

C.5.7 It is desirable but not required if the Contractor has capabilities in the following areas:

C.5.7.1 MADYMO or another lumped mass modeling software package: These packages will be used when a simplified lumped mass model can be used to save computing time and support parametric analysis

C.5.7.2 Restraint Design: Ability to design or modify a restraint system to consider all loads applied to the dummy during a test.

C.5.7.3 Suspension characterization testing: This provides data to support realistic simulation of vehicle pitching and handling.

C.5.5.4 Component testing, sled testing, pendulum tests: To provide test data to evaluate subsystem models.

C.5.8 The Contractor shall have appropriate staff and access to software and facilities to conduct large vehicle crash simulations using existing finite element models developed for NHTSA. Software capabilities must be able to directly input and output models in LS-Dyna format.

C.5.9 The Contractor shall be capable of vehicle measurement, scanning, disassembly and digitization. The vehicle scanning capability shall capable of scanning and digitizing complex vehicle shapes within an accuracy of 1mm. The scanning data shall be converted to surface data and/or optimized mesh geometry suitable for CAD modeling and crash simulation. Final model assembly should include capabilities for evaluating mesh quality, part interference, and contact definition.

C.5.10 The Contractor shall have the capability or access to computer facilities capable of running a full vehicle LS-Dyna crash simulation within 16 hours, assume 2 million elements running for a 150 milli-seconds(ms) simulation time.

C.5.11 The Contractor shall have the capability for sample preparation and testing to characterize material response under crash or impact loading conditions. This capability includes the development of simulation input parameters from reference and/or test results. The material parameters should be appropriate for common automotive materials and loading conditions.

C.5.12 The Contractor shall have clearly defined procedures and standards for comparing crash simulation results against staged test results. The Contractor shall have a methodology to document objective measure(s) to evaluate the fidelity of simulation results.

C.5.13 The Contractor shall have the capability for conducting simulations and interpreting the results from human body models. The Contractor shall have the capability to work with these models and evaluate the simulation outputs to evaluate occupant injury potential.

C.5.14 The Contractor shall have the capability to conduct finite element occupant interior simulations including dummy positioning, modeling restraint performance, air bag folding and deployment. This capability should include finite element pre-simulation for seat squashing and belt routing.

C.5.15 The Contractor shall have the capability to conduct parameter and optimization studies using finite element models. Parameter studies will involve evaluating safety implications for a range of parametric values. Optimization studies will be similar but will include multiple parameters and the development of appropriate objective functions to enhance the safety performance.

(End of Section C)